

CHAPTER – 01

INTRODUCTION

1.1 – Project Overview

Introduction

Electricity is a fundamental resource that powers homes, businesses, industries, and modern technologies. As global energy demand continues to rise due to population growth, urbanization, and increased reliance on electronic devices, understanding electricity consumption patterns has become essential. This project explores how electricity is consumed across different sectors, identifies factors influencing usage, and examines opportunities for improving energy efficiency and sustainability.

Project Purpose

The purpose of this project is to analyze electricity consumption trends and patterns, understand consumer behavior, and identify practical strategies for reducing energy waste. By studying consumption data, the project aims to provide insights that support informed decision-making by consumers, utility providers, policymakers, and businesses.

Objectives

- Analyze electricity consumption patterns in residential, commercial, and industrial sectors.
- Identify peak and off-peak usage periods.
- Examine factors affecting electricity demand, such as weather conditions, appliance usage, population growth, and economic activity.
- Evaluate the impact of energy-efficient technologies and renewable energy adoption.
- Provide recommendations for optimizing electricity consumption and reducing energy costs.